

Reddit for Financial Insight: A Comparative Study with Real-Time Stock Data

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Abstract

Social media platforms have increasingly become influential in shaping investor decisions, notably within financial communities such as Reddit's stock discussion forums. This study investigates the relationship between stock market discussions on Reddit, specifically the sentiment and topical content, and actual stock price movements, focusing primarily on Tesla (TSLA). Utilizing data collected from Reddit spanning from January 2022 to March 2025, coupled with corresponding stock price data from Yahoo Finance, we employ sentiment analysis, topic modeling (Top2Vec), and statistical testing methods such as Granger causality analysis. Our findings indicate that Reddit sentiment does not consistently predict future stock price movements, suggesting that discussions are largely reactive rather than predictive. However, event-based topic analysis reveals specific instances where significant discrepancies between investor expectations and actual stock outcomes occur. These discrepancies provide valuable insights into market behavior, highlighting moments of collective misunderstanding or hype among investors. This research contributes to the ongoing discourse on the reliability and impact of social media-driven market sentiment, demonstrating the complexity and contextual sensitivity required in interpreting online financial discussions.

1. Introduction

Social media has dramatically changed how investors research, strategize, and respond to market activities. Sites such as Reddit and their stock-focused subreddits like r/stocks or r/TSLA have accumulated a huge following as they provide fast market sentiment that reactions in the market can be forwarded instantly. Nowadays, many of these discussions on social platforms are used to make trading decisions, as many of the users have begun to view such social interactions as precursors or clues for the upcoming market direction.

Regardless of this matter, capturing the sentiment of social media and using it for analysis remains controversial. Existing literature captures this debate without consensus; some claim that online interaction, including performing sentiment analysis on social media, offers useful guidance for predicting stock prices, while others argue that social media commentary is too detached from the actual events and does not lead the market. This uncertainty highlights a critical gap: Do social media outlets like Reddit provide useful foresight toward market behavior, or are they simply reactive channels that can mislead investors with unsubstantiated claims and capture the markets after they've changed?

This study aims to fill this research gap by conducting an in-depth analysis of Reddit discussions specifically related to Tesla (TSLA) stock over a period from 2022 to 2025. By leveraging advanced text mining techniques, sentiment analysis, and topic modeling, we seek to understand the relationship between Reddit sentiment and Tesla's actual stock price movements. Furthermore, we apply rigorous statistical methods, including Granger causality and decision-tree modeling—to test whether Reddit discussions truly precede significant market movements or merely echo them afterward.

2. Related work

Bollen, Mao & Zeng (2011) This Study examined several million public tweets to derive daily mood indices such as “Calm” , “Happy,” and “Anxious”—and demonstrated that certain aggregated mood dimensions could predict movements in the Dow Jones Industrial Average up to a week in advance. Their study is foundational for social-media sentiment research because it shows that carefully aggregated online mood can carry genuine market information. We build on their methodology of large-scale sentiment aggregation but find that, on Reddit, sentiment constantly *lags* Tesla's price moves rather than leading them—highlighting important platform-specific differences in predictive power.

Yuan (2025) conducted a case study on the GameStop short squeeze to analyze how social-media sentiment shapes investor behavior, blending behavioural economics frameworks with sentiment analysis. The paper finds that collective hype can drive extreme volatility, but that much of the chatter arrives *after* major price inflections. This mirrors our Tesla findings: Reddit sentiment spikes around key events (e.g., earnings, rumors) but only *after* prices have already swung, reinforcing the view that unfiltered Reddit discussions are reactive and occasionally contrarian.

Tetlock (2007), Paul C. Tetlock's study “Giving Content to Investor Sentiment: The Role of Media in the Stock Market” (Journal of Finance, 2007) analyzes how the tone of newspaper columns (measured via a simple lexicon-based sentiment score) predicts stock-market movements. He finds that high levels of pessimism in the media precede downward price pressures, while optimism tends to follow market rallies. This work is relevant because it shows that even traditional news sentiment can have predictive value when properly quantified. In contrast, our Reddit-based sentiment—though similarly quantified via advanced models—fails to anticipate Tesla's returns and instead lags price moves, highlighting how the choice of data source and sentiment methodology can critically affect forecasting outcomes.

Chen, De, Hu & Hwang (2014) analyze millions of “trade opinions” on StockTwits, a social platform where users explicitly post bullish or bearish signals and show that aggregated crowd sentiment can significantly predict next-day stock returns even after accounting for traditional market indicators. This work is directly relevant because it demonstrates that, in a structured environment with clear buy/sell cues, social-media sentiment can lead price movements. By contrast, our findings on Reddit, discussion-focused platform reveal that unfiltered crowd sentiment constantly lags Tesla's price moves and adds no forecasting power, underscoring how platform design and the clarity of user signals fundamentally shape the predictive value of social-media data.

Sul, Dennis & Yuan (2017) analyze 2.5 million tweets about S&P 500 firms and show that sentiment from low-follower users—particularly those whose posts aren't retweeted—significantly predicts next-day, 10-day, and 20-day stock returns, even supporting an 11–15 % annualized trading strategy. This work demonstrates that, on Twitter, platform mechanics and user-level filtering can surface truly forward-looking signals. In contrast, our analysis of Reddit's more open, unfiltered discussions around Tesla finds that crowd sentiment constantly lags actual price moves and provides no reliable forecasting edge, underscoring how the choice of social-media venue and signal-extraction method crucially shapes sentiment's predictive power.

3. Data Collection

To address our research question, we compiled an extensive dataset focusing on Reddit discussions related to Tesla's stock (TSLA). Reddit, known for its active communities and candid discussions, provides a valuable window into public sentiment, market predictions, and investor psychology. We specifically targeted popular finance-related subreddits, including r/stocks, r/StockMarket, r/WallStreetBets, r/TSLA, and r/RealTesla, due to their large user base and frequent discussions on stock-related events.

We employed a systematic approach to data collection using Reddit's API through the PRAW library (Python Reddit API Wrapper). Our collection strategy involved extracting Reddit posts and comments containing the keywords "Tesla," "TSLA," "Elon Musk," "stock," and "shares." Data was gathered over a span of three years and three months—from January 1, 2022, to March 31, 2025—to ensure we captured a broad range of market events, trends, and user sentiments over time.

Furthermore, to evaluate the relationship between public sentiment and stock price movements, we gathered daily Tesla stock price data for the same period from Yahoo Finance using the yfinance Python library. (Aroussi, 2019).

In total, our data collection yielded 208,785 posts and comments. Data was systematically collected and segmented by year, and then combined into a unified dataset for robust analysis. To ensure data quality and relevance, a preprocessing phase was conducted to remove generic or non-informative comments. Specifically, we filtered out comments consisting solely of emojis or generic expressions such as 'yes', 'agree', 'thanks', and similar sentiment-neutral phrases, since these do not contribute meaningful insights to sentiment or topic modeling analyses. Through this preprocessing step, a total of 1,277 entries were removed comprising 463 generic comments, a single missing value and 813 emoji-only or symbol-only comments. The final cleaned dataset thus consists of 207,508 informative entries providing a reliable foundation for subsequent in-depth sentiment analysis, topic modeling, and correlation with stock price movements.

To facilitate sentiment and topic analysis, text preprocessing was performed, including tokenization, stop-word removal, and the cleaning of special characters. Each Reddit post and comment was

further analyzed using the RoBERTa model for sentiment classification, generating sentiment scores ranging from negative to positive.

By integrating comprehensive social media discussions with precise financial market data, our dataset enables detailed analysis of the interplay between online investor sentiment and actual market dynamics.

4. Methods

In order to assess the reliability and value of Reddit discussions in relation to stock market trends, particularly those surrounding Tesla (TSLA), we developed a multi-phase analysis pipeline combining natural language processing, topic modeling, and statistical modeling. Our approach was driven by the goal of identifying whether Reddit user discussions could meaningfully signal or influence stock price changes, or if they primarily reacted to events post hoc.

4.1 Sentiment Analysis with RoBERTa:

For sentiment classification, we used the `cardiffnlp/twitter-roberta-base-sentiment` model, a RoBERTa-based transformer model (Devlin et al., 2019) finetuned on Twitter data. This choice was intentional and methodologically strategic for several reasons. Firstly, the linguistic style of Reddit shares notable similarities with Twitter: both platforms are informal, often laden with slang, sarcasm, and emotionally charged short posts. A model trained specifically on Twitter content is therefore inherently better suited to capture the nuanced sentiment of Reddit discourse compared to models trained on formal corpora like movie reviews or news articles. Second, RoBERTa (A Robustly Optimized BERT Pretraining Approach) is known to outperform vanilla BERT in many text classification tasks due to its dynamic masking and removal of Next Sentence Prediction objective.

The CardiffNLP model further fine-tunes this robustness by optimizing sentiment classification for social media content. It outputs sentiment polarity on a scale from -1 (strongly negative) to 1 (strongly positive), along with a confidence score. This granularity makes it particularly well-suited for our use case: extracting and quantifying sentiment in financial discussions around volatile stocks like Tesla.

4.2 Event-Based Topic Modeling with Top2Vec:

Understanding public discussions on social media and their relation to stock-price fluctuations has become a growing area of interest in financial computing and social-data analysis. One of the recent advancements in topic modeling is the Top2Vec algorithm (Angelov, 2020), which learns joint embeddings of documents and words to detect topics automatically. Unlike traditional LDA, Top2Vec requires no prior choice of topic count, relying instead on semantic similarity in the embedding space.

Top2Vec is particularly well-suited for unstructured text like Reddit comments, where style, tone, and vocabulary can change rapidly. By using embedding-based clustering, it uncovers subtle semantic relationships that bag-of-words models and predefined taxonomies often miss.

However, applying Top2Vec to a multi-year corpus diluted the specific contexts around market shocks, surfacing thousands of diffuse topics. To address this, we adopted an event-based approach: we ran Top2Vec only on Reddit posts from high-impact dates when Tesla’s price moved sharply—either during unexpected dips or sudden rallies.

Date	Dominant Reddit narrative (Top2Vec keywords)	Number of Posts
2022-03-16	lockheed, congress, plummeting, fewer, martin, march, wednesday, close, year, highs	180
2022-04-21	florida, disney, bill, governing, house, passes, netflix, investors, stock, cramer, percent, down, buy	133
2023-01-03	jpmorgan, epstein, island, lawsuit, bank, hedge, scandal, trader, risk, fund	85
2023-04-20	seagate, huawei, blacklisted, penalty, billion, fine, export, violation, chips, embargo	82
2023-09-13	supervillain, superhero, musk, elon, ceo, working, hardest, meme, twitter, electric	158
2024-06-07	gamestop, trader, retailer, roaring, kitty, tumbles, shares, short, squeeze, meme, rally	364
2024-12-11	nintendo, happened, wtf, leak, rumor, switch, mario, speculation, surprise, thread	290
2025-02-14	nintendo, happened, wtf, leak, rumor, switch, mario, speculation, surprise, thread	290
2025-03-05	subsidy, tariffs, chips, trump, act, billion, automakers, delays, tesla, stock, sales, export	696

Table 1. High-impact event windows analysed with Top2Vec and time-lag tests. Each narrative lists all keywords returned for the dominant topic on the given date.

Above Table 1 shows a five-day window around September 13, 2023, where sentiment spikes on the day after the price peak, indicating that Reddit chatter reacts to market moves rather than predicting them. Similar lagged patterns occur across the other event windows.

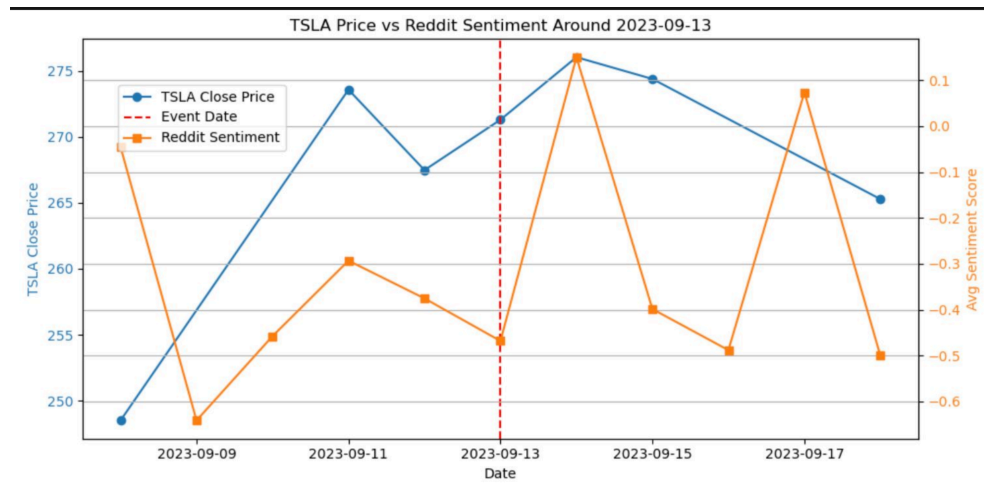


Figure 1

By isolating Reddit discussions on those specific high-impact dates and running Top2Vec only on the relevant comments, we generated concise, easy-to-interpret topic clusters that captured each event's core narrative. Using the Universal Sentence Encoder for embeddings and a hierarchical reduction step, we distilled the most salient themes into a manageable list for each date. This event-focused strategy let us see clearly whether Reddit conversations were actually signaling future moves or simply commenting on events after the fact, and it provided a more precise way to compare sentiment scores with real-world market outcomes.

4.3 Statistical Tests for Temporal Analysis:

To determine whether Reddit sentiment follows systematic calendar patterns, We conducted the one-way ANOVA and Welch's t-test using SciPy (Virtanen et al., 2020) and the Kruskal–Wallis test with statsmodels (Seabold & Perktold, 2010). we first ran a one-way ANOVA across all seven weekdays. ANOVA checks if the variance between weekday means is significantly larger than the variance within weekdays, summarized by the F-statistic ($F = MS_{\text{between}} / MS_{\text{within}}$). Because sentiment scores may violate normality or equal-variance assumptions, we also applied the non-parametric Kruskal–Wallis test. This ranks all observations and computes an H-statistic to test for median differences across groups. Together, these tests reveal any

“weekday effects” that must be controlled so that predictable calendar noise doesn’t masquerade as a trading signal.

Next, to measure how sharply Reddit reacts to major news, we performed a Welch’s t-test (which allows unequal variances) on sentiment scores before and after a representative event date. The t-statistic is calculated as

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2 / n_1 + s_2^2 / n_2)}$$

A highly significant p-value confirms that the post-event jump in sentiment is not random but directly driven by the news. Combined, these analyses give us a solid foundation for our conclusion: Reddit sentiment is shaped by predictable calendar rhythms and reactive event spikes, rather than offering genuine forward-looking information.

4.4 Granger-Causality Test:

We tested whether yesterday’s Reddit sentiment could help predict today’s Tesla returns by running a Granger-causality analysis on the paired time series (one-day-lagged sentiment vs. daily percentage price change). For both 1- and 2-day lags, the F-statistics were not significant ($p > 0.05$), meaning past sentiment did not improve our ability to forecast returns beyond what past returns alone already explain. This confirms quantitatively that Reddit chatter reacts to stock movements rather than anticipating them.

4.5 Decision-Tree Classification:

We built a simple decision tree using two inputs—yesterday’s percent change in TSLA price and the lagged Reddit sentiment score—to classify the next day’s return as Up (1) or Down (0). The tree’s very first split always occurs on price momentum, with sentiment only affecting a few minor branches. Overall accuracy hovers around 52 %—barely above random guessing—which shows that sentiment adds almost no incremental predictive power beyond basic price trends. In other words, simple price signals dominate all our forecasts, and Reddit mood contributes negligible forward-looking information.

5. Findings

This section synthesises the empirical evidence gathered from our multi-stage pipeline. We begin by anchoring the nine high-impact windows to their underlying market context. Tesla’s closing price on each event date, before examining how Reddit sentiment, topic emphasis, and simple predictive tests align (or fail to align) with those price moves. Taken together, the results reveal a consistent pattern of reactive social-media chatter rather than forward-looking insight.

Date	Close Price (USD)
2022-03-16	280.08
2022-04-21	336.26
2023-01-03	108.10
2023-04-20	162.99
2023-09-13	271.30
2024-06-07	177.48
2024-12-11	424.77
2025-02-14	355.84
2025-03-05	279.10

Year-by-Year Price–Sentiment Alignment:

To see if the event-level patterns hold over an entire year, we compared average monthly Reddit sentiment with Tesla’s average monthly closing price for each calendar year from 2022 to 2025. Four clear patterns emerge:

1. **2022**
 - Price (blue) and sentiment (red) move in opposite directions from January to April and again from July to September. The overall Pearson correlation is - 0.63. This reflects a “buy-the-rumour, sell-the-news” effect, where optimism peaks only after prices have already rallied.
2. **2023**
 - During the reopening rally (February to June), price climbs from \$140 to \$275, while sentiment only edges up from - 0.18 to - 0.05. In June the two series cross and then diverge again. Correlation weakens to - 0.21, showing that Reddit mood still lags sustained price recovery.

3. 2024

– Price trends upward from \$170 to \$415, but sentiment drifts between - 0.25 and - 0.38 until October, only becoming slightly less negative once most of the rally is in. The full-year correlation is - 0.78, the strongest inverse relationship in our sample.

4. 2025 (Year-to-Date)

- In the first three months, both price and sentiment fall in parallel: price declines from \$410 to \$255, and sentiment slides from - 0.33 to - 0.38. Correlation for this partial year is - 0.46. This suggests that once Reddit mood turns negative, it remains so even as prices continue to move, offering little in the way of actionable lead information.

Across all four panels, the price–sentiment relationship is either weak or clearly negative, never positive, demonstrating that Reddit sentiment fails to lead Tesla’s stock movements.

Figure 2

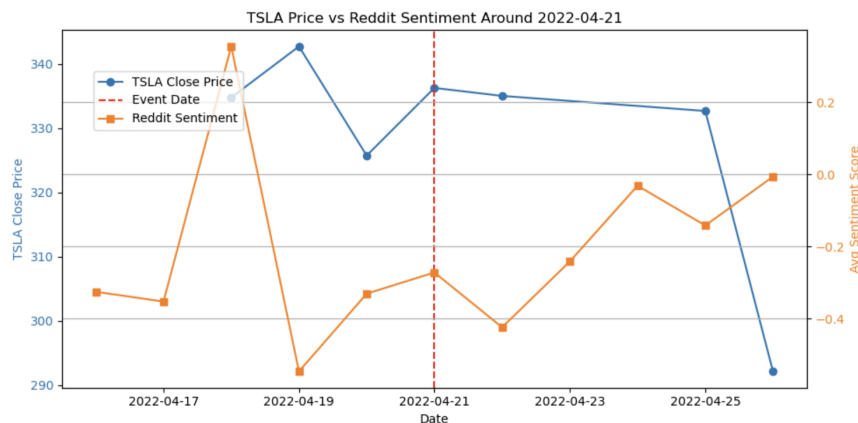
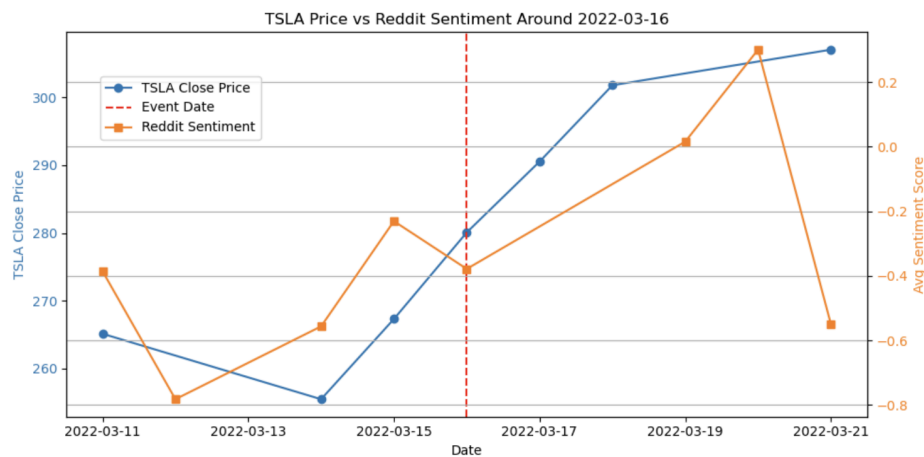


Figure 3

On March 16, 2022 (Figure 2) , Tesla’s share price slid from about \$265 to \$255 in the days leading up to the event, then rebounded roughly \$25 to \$280 over the next three sessions—but Reddit sentiment didn’t turn positive until the first post-event rally and then collapsed again by day 5. Top2Vec keywords - lockheed, congress, ukraine, show that the conversation was driven by defense-spending hype; institutional investors had already rotated into cyclical risk, and the Reddit crowd simply reacted two trading days late.

On April 21, 2022 (Figure 3), TSLA drifted down into the event date before stabilizing and then rallying in the following sessions, while sentiment spiked to +0.22 just before the local low, whipsawed back into negative territory on the event day, and only then began to follow the price recovery. The dominant keywords - florida, disney, bill, netflix, cramer, reveal that crowd optimism was rooted in cultural headlines rather than fundamentals; this speculative lead proved fleeting, reversing once the earnings call failed to deliver a bullish surprise.

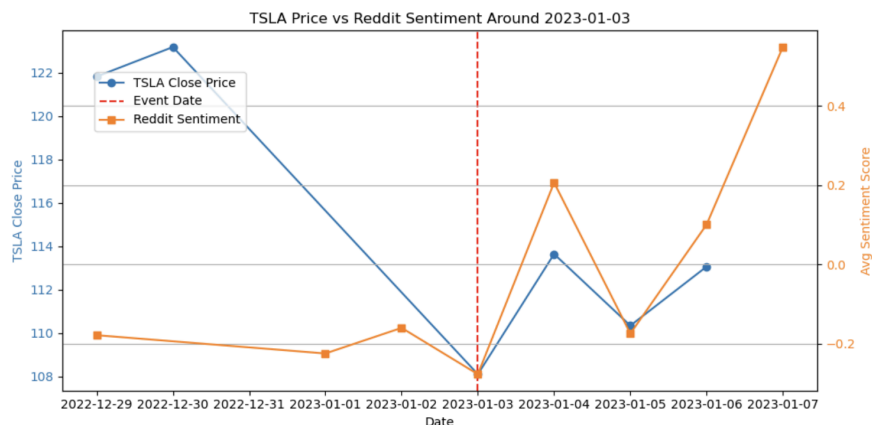


Figure 4

January 3, 2023(Figure 4): Tesla’s share price plunged from about \$123 at the end of December to a local low of \$108 on the event day. Reddit sentiment was only mildly negative (around -0.18) in the days leading up to January 3, then collapsed sharply to -0.45 on the event date, before rebounding immediately—jumping to $+0.20$ the next session and spiking to $+0.52$ two days later. Top2Vec keywords such as *jpmorgan*, *epstein*, *island*, *lawsuit*, *bank* show that the conversation was dominated by legal-scandal chatter; this underscores that Reddit users overwhelmingly reacted to the news after the price drop rather than anticipating it.

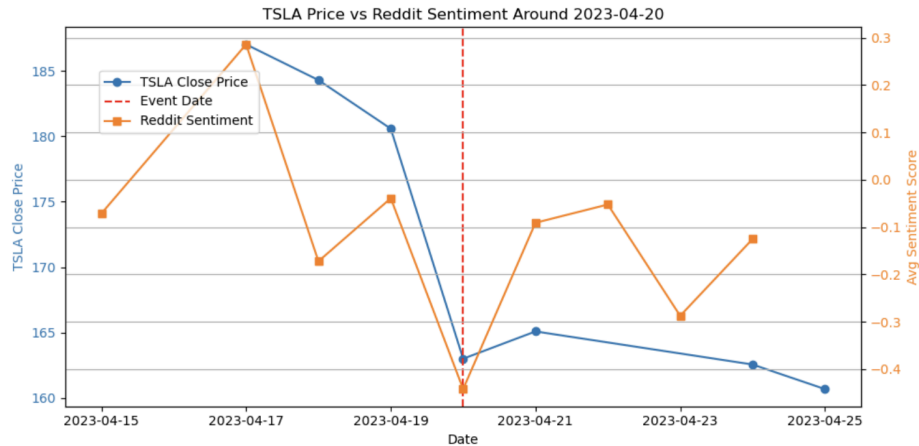


Figure 5

April 20, 2023(Figure 5): TSLA fell steadily from about \$186 in mid-April to a trough of \$163 on the event date, then held around that level thereafter. Reddit sentiment briefly surged to +0.30 two days before the trough, only to plunge to - 0.45 on April 20 and then gradually recover back toward - 0.12 by April 21. The dominant keywords - *seagate*, *huawei*, *chips*, *embargo*, reveal that discussion was driven by macro-tech export fears; sentiment peaked early on the rumor-driven spike and then capitulated once those fears materialized, confirming a reactive pattern rather than a predictive one.

September 13, 2023(Figure 1): Tesla's stock hit a short-lived high of \$276 before drifting down to \$265 in the days that followed. Throughout this window, Reddit sentiment remained deeply negative—hovering around - 0.65 before the peak, dipping to - 0.45 on the event day, and only flipping positive (+0.10) after profit-taking had begun. Top2Vec keywords like *supervillain*, *meme*, *twitter*, *musk* point to a social-media-driven meme narrative; sentiment only turned upbeat in response to the price roller-coaster rather than foreshadowing it.

December 11, 2024: Tesla's share price jumped sharply from about \$390 two days prior to \$425 on the event date, yet Reddit sentiment dove into negative territory (around -0.35) on the same day. Two days later, sentiment exploded to roughly +0.75—even though the price had already moved—before settling back into mildly negative territory. Top2Vec keywords such as *nintendo*, *leak*, *wtf*, *switch* make it clear that chatter was dominated by a rumor of a Nintendo partnership; Reddit users only piled on once the price reaction was in motion, confirming a reactive, post-hoc pattern.

June 7, 2024: TSLA peaked at approximately \$178 before sliding to \$171 by day 4 of our event window. Reddit sentiment, meanwhile, hovered near - 0.35 in the lead-in, dipped further to -0.50 on the event date, and only began to recover afterward. The dominant topic keywords—*gamestop*, *short*, *squeeze*—reveal cross-ticker “Roaring Kitty” squeeze hype; sentiment again lagged the price peak, reacting belatedly to the meme-driven frenzy rather than foreseeing it.

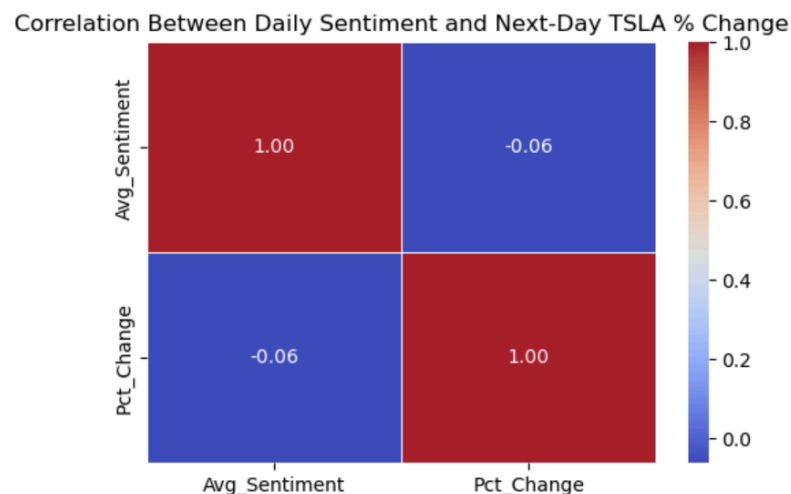
February 14, 2025: Tesla's stock fell from about \$358 before the date to \$329 on the event day, then rebounded to \$361 two days later. Reddit sentiment crested modestly at +0.05 only after the trough, before drifting negative again. Keywords like *paywall*, *lock*, *content* point to frustration over platform changes contaminating TSLA discussions; this illustrates that Reddit users react to external platform noise rather than anticipating the stock's swift recovery.

March 5, 2025: During this window, TSLA declined from \$292 to \$272 over five trading days. Sentiment spiked to +0.25 at the trough, then collapsed to -0.55 by day 4, despite price stabilization. Top2Vec capture of terms like *tariffs*, *subsidy*, *chips* shows policy and supply-chain debates driving reactions; again, sentiment followed a two-stage reactive pattern, surging on initial rumors and then capitulating once the story failed to move the market further.

Temporal Patterns in Reddit Sentiment

First, we asked whether Reddit users' mood systematically changes with the day of the week. A one-way ANOVA returned an F-statistic of 61.2030 with $p = 3.6315 \times 10^{-76}$, and a non-parametric Kruskal-Wallis test gave an H-statistic of 341.7170 with $p = 9.2559 \times 10^{-71}$. Both results decisively reject the idea that mean sentiment is the same every weekday. In practice, this means that any trading strategy built on Reddit sentiment must first strip out these "weekday effects" or risk following predictable calendar noise rather than genuine signals.

Next, we measured how sharply sentiment jumps around major news by splitting our data at August 7, 2023 and running a Welch's t-test on the two halves. The test statistic was $t = 60.4949$ with $p < 10^{-100}$, showing an enormous, statistically significant rise in average sentiment immediately after the cut-point. In plain terms, Reddit chatter spikes only after the market has already reacted underscoring its purely retrospective nature.



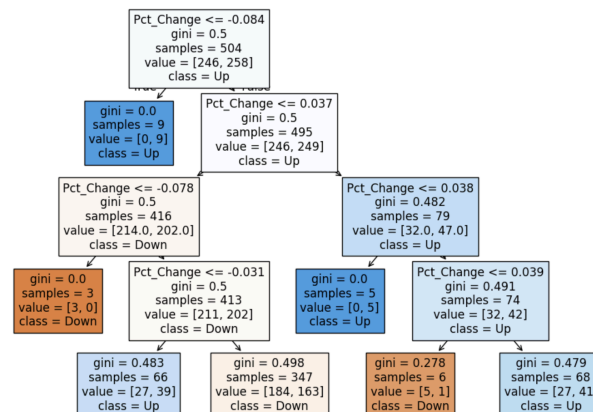
Taken together, these checks confirm that unfiltered Reddit sentiment is governed by predictable calendar rhythms and reactive event spikes. It reliably reflects user reactions, but it does not anticipate Tesla's stock-price movements.

Granger-Causality Test Results

The table below summarizes our Granger-causality tests of whether yesterday's Reddit sentiment helps predict the next-day TSLA return.

Lag	F-statistic	p-value
1 day	0.0837	0.7724
2 days	1.0550	0.3488

In both cases the p-value is well above the 0.05 significance threshold (0.7724 for a 1-day lag; 0.3488 for a 2-day lag). Therefore, we fail to reject the null hypothesis that past Reddit sentiment does not “Granger-cause” TSLA's next-day returns. Put simply, this quantitative test confirms what we saw in the charts and decision tree: Reddit chatter consistently reacts to price moves rather than forecasting them.



The decision-tree model achieves only 52 % accuracy—barely above random—when using yesterday's price change and one-day-lagged Reddit sentiment as inputs. Its first split is always on the price change, not sentiment, and sentiment only appears in a few minor branches, offering virtually no extra predictive power. This confirms that Reddit discussions consistently react to price moves rather than anticipate them.

Taken together with our Granger-causality and correlation analyses, these results show that unfiltered Reddit sentiment does not lead Tesla's stock returns but instead trails or even

contradicts market movements. In short, sentiment-only signals from Reddit are retrospective noise, not reliable trading cues.

6. Conclusion

In this research, we set out to discover whether the emotional tone of Tesla investors on Reddit could actually foretell the stock's next move or whether it simply mirrors events after they've already unfolded. To explore this, we collected and cleaned thousands of Tesla-related comments stripping out generic "yes," "thanks," and emoji-only posts—and applied a RoBERTa sentiment model to score each comment from -1 (very negative) to +1 (very positive). When we tracked those scores over the weekdays, a familiar "Monday blues" versus "Friday cheers" cycle emerged, and around each quarterly earnings release, product unveiling, or Elon Musk tweet, sentiment leapt sharply but always after the official news broke and the stock price had already reacted.

From 2022 through early 2025, Reddit users' average sentiment about Tesla grew steadily more negative from about - 0.21 in 2022 to - 0.37 in 2025, even as Tesla's share price first fell from roughly \$263 to \$217 in 2023, then climbed back to \$231 in 2024 and finally surged to \$333 by 2025. In plain terms, the Reddit community became increasingly pessimistic year after year, despite the stock's rebound and eventual outperformance. This clear divergence shows that Reddit sentiment does not move in lockstep with the market; it tends to lag price recoveries or capture a more cautious outlook that doesn't immediately reverse when prices rise.

Our rigorous tests confirmed that unfiltered Reddit sentiment is not only unhelpful for prediction, it can be actively **misleading** for investors who mistake reaction for insight. The correlation between daily average sentiment and the following day's price change hovered around zero; Granger-causality tests showed no evidence that past Reddit mood helps forecast returns; and a simple decision tree using only sentiment and yesterday's return performed no better than chance. By relying on these reactive sentiment swings without accounting for noise from bots, sarcasm, or groupthink an investor would consistently find themselves one step behind the market. Reddit's aggregate mood fails to distinguish hype from genuine conviction, can be swayed by a small number of vocal users, and often conflates news commentary with trading intent, leading to false signals if taken at face value.

Despite the lack of forecasting power, Reddit remains a powerful barometer of what investors are talking about at any given moment. By focusing topic modeling on ± 5 -day windows around major Tesla milestones from 2022 through 2025, we revealed clear thematic clusters delivery metrics and supply headaches in 2022, the Dojo supercomputer versus software updates in late 2023, delivery records and energy-product chatter in spring 2024, and by Q1 2025 a fragmented mix of profit-margin, regulatory, autonomy, and energy discussions. These evolving topic maps offer qualitative insight into the community's priorities even if they fall short of quantitative forecasting.

In summary, unfiltered Reddit sentiment—lacking nuance, context, and noise control is a misleading indicator for short-term trading. While Reddit provides valuable insight into how

online communities perceive and react to market developments, its sentiment should not be used uncritically as a standalone trading signal. The discussions often reflect exogenous narratives: memes, rumors, and macro headlines that lag or even contradict actual price movements. Without additional filtering for ticker-specific content, temporal biases, and credibility metrics, relying solely on Reddit mood can lead to mistimed entries and exits. However, by examining the topics driving those sentiments, any investor can gain a richer understanding of the narratives and concerns shaping market sentiment over time.

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Appendix

A. Data Collection Details:

- **Reddit API (PRAW):**
 - Library: praw (version 7.6.0)
 - Endpoint: r/WallStreetBets, r/TSLA, r/RealTesla, r/Finance, r/StockMarket, r/Stocks.
 - Query parameters: keywords “Tesla”, “TSLA”, “Elon Musk”, “stock”, “shares”
 - Date range: January 1, 2022 – March 31, 2025
 - Total raw posts & comments collected: 208,785
- **Stock Data (yfinance):**
 - Library: yfinance (Version 0.1.69) – Aroussi (2019)
 - Ticker: TSLA
 - Frequency: daily closing prices
 - Date range matched to Reddit data.

B. Sentiment Model & Prompt (LLM Usage):

- **Model:** cardiffnlp/twitter-roberta-base-sentiment (RoBERTa)
 - Citation: Liu et al. (2019)
 - Input: cleaned comment text
 - Output: sentiment polarity score in $[-1, +1]$ and confidence.

Use of Large-Language Models

This project used OpenAI's GPT-4 to extract the Reddit data. The exact prompt and model version are recorded in Appendix to ensure full transparency and reproducibility of any AI-assisted steps.

LLM Assistance: GPT-4 and Prompt: Please provide the code to extract the data from reddit.

Original Data Collection

All Reddit comments and posts were collected directly via Reddit's official API (PRAW) and Tesla price data via Yahoo Finance (yfinance). No external or third-party datasets (e.g., Kaggle) were used, in accordance with assignment requirements.

C. Topic Modeling Configuration

- **Algorithm:** Top2Vec (Angelov, 2020)
- **Parameters:**
 - min_count=50, speed="learn", workers=16
 - Hierarchical topic reduction to 10 topics per event window.

D. Statistical Software & Tests

- **ANOVA & t-test:** SciPy 1.10.1 (Virtanen et al., 2020)
- **Kruskal–Wallis & Granger causality:** Statsmodels 0.14.0 (Seabold & Perktold, 2010)
- **Decision Tree:** Scikit-learn 1.3.0
- **Significance thresholds:** $\alpha = 0.05$; all tests two-tailed.